

Yujin Song

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EDUCATION

Simon Fraser University, Faculty of Applied Science

Bachelor's degree in Computing Science | Minor in Mathematics

Burnaby, BC

Expected May 2027

- **Relevant Coursework:** Data Structures and Algorithms, Probability and Statistics, Linear Algebra, Linear Optimization, System Programming, Introduction to Artificial Intelligence

Fraser International College

Major in Computing Science

Burnaby, BC

2023 September - 2024 August

- **Relevant Coursework:** CalculusI, CalculusII
- **Awards:** Nominated student for receiving a GPA over 3.67

TECHNICAL SKILLS

- **Programming Language:** Python, C++, C, HTML, CSS, Javascript, Typescript, Ruby, Assembly
- **Framework:** FastAPI, Express.js, React.js, Ruby on Rails, TailwindCSS
- **Other Skills:** SQLite, MongoDB, Node.js, Git, Docker, Bash Scripting, Reversing, Cross-Site Scripting, OpenCV, RestAPI, Vite

PROJECTS

Aether (Association of Korean-Canadian Scientists and Engineers, SFU) | Python, OpenCV, Bash, Docker

Burnaby, BC

Software Team

Mar 2026 – August 2026

- Produced **ROS2** and **Gazebo** documentation to support **reducing knowledge gaps** across the software team.
- **Authored** and maintained **pytest.sh**, a Bash script that improves team **development workflow consistency**.
- Implemented a **hand-grabbing detection feature** for the Computer Vision node using **OpenCV**, measuring and normalizing fingertip-to-palm distances with separate open/close thresholds to drive gripper control while preventing signal flickering.

Schedule Mate | MongoDB, Express.js, Node.js

Full Stack Development

May 2026 – July 2026

- Built a **full-stack timetable-sharing web app** with a React (Vite) frontend and **Node.js/Express + MongoDB** backend, replacing screenshot-based schedule sharing with **RESTful APIs** for schedule CRUD and a friend system to compare timetables side by side.
- Adapted and integrated a JWT-based authentication flow into the app, and added Upstash Redis rate limiting to protect API endpoints.
- Integrated a public university course API through a backend proxy layer, normalizing live course/section data into schedule blocks users can add directly to their timetable.

Oh My Graph | C#, Unity

Burnaby, BC

Game Developer | 3rd Place, Best Game

Sep2026

- Developed a **2D educational runner game** in **C# and Unity** where players complete mathematical functions to generate paths and guide a character through each level.
- Implemented dynamic **piecewise-function generation**, keyboard-based answer input, real-time graph visualization, and progressively increasing gameplay speed.
- Collaborated in a **two-person team** to design and deliver a playable game within a two-day game jam, earning **3rd place in the Best Game category**.

Simple Shell | C, Linux

CMPT 201 System Programming Project

June 2026

- Implemented a **Unix-like shell** in **C** supporting process creation, foreground/background execution, and built-in commands.
- Designed command history and process management using `fork()`, `execvp()`, and `waitpid()`, including zombie process cleanup.

LEADERSHIP & CLUB & LAB EXPERIENCE

SFU Cybersecurity Club

Burnaby, BC

Co-Founder, President

Jan 2025 – Present

- **Established** and **structured organization** with systematic procedures and frameworks, **leading 8+** executive members as a co-founder and the initial president of the club.
- **Organized diverse cybersecurity activities**, including CTFs and workshops, to create an environment where members can first dive into cybersecurity.
- Contributed leading **50+** undergraduate students to the cybersecurity community by bringing together individuals interested in the field.

SFU Ascension Robotics | C++, OpenCV, Linear Optimization, Linear Algebra

Surrey, BC

Computer Vision Team Member

Jan 2026 – Present

- **Co-developed** a mathematically grounded `findRobotCenter` function using **linear algebra** and geometric constraints, and independently implemented a `drawSelectedPosition` function using **OpenCV** to enable accurate aiming practice on target panels.
- Implemented a teammate-designed yaw estimation function by integrating reusable components and **translating mathematical formulations** into working **C++** code.

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- Supported the Business Team in planning a large-scale event, contributing venue recommendations that **reduced** projected costs by ~\$700.

Math Student Union SFSS Representative

Burnaby, BC

Non-Executive Councillor

May 2026 – Present

- Represent the **Mathematics Student Union (MSU)** at **Simon Fraser Student Society (SFSS) Council** meetings as a **non-executive councillor**.
- **Communicate SFSS council discussions**, policies, and initiatives to the MSU executive team, serving as a liaison between the two organizations.

MARS Lab, Simon Fraser University | Python, ROS 2

Burnaby, BC

Volunteer Student Researcher

Jul 2026 – Present

- Studying **Hamilton–Jacobi reachability** and **Control Barrier Function controllers** for safety-critical robotics and future **sim-to-real deployment**.
- Gained hands-on experience with **TurtleBot3**, **Unitree**, and **AgileX** simulation platforms by setting up and running environments with **ROS 2**, **MuJoCo**, and **RViz**.

WORK EXPERIENCE

English Lecturer Assistant and Manager

Goyang-si, KR

ETOOS (Korean Private Academy) | Part-time

Mar 2023 – Jul 2023

- Managed attendance and assignment tracking for **30+** students using an **internal platform**, maintaining clear communication with both students and parents.
- **Cooperated** on a score-analysis spreadsheet for performance tracking using Excel ensuring **clear communication** with coworkers.

EXTRACURRICULAR EXPERIENCE

Clash of Equations (Math Competition)

Burnaby, BC

Team Leader

Mar 2026

- Led a 3-person team across three distinct math domains (Calculus, Linear Algebra, Number Theory), coordinating strategy and communication prior to competition.
- Achieved **full marks in Calculus** and partial marks in **Linear Algebra**, contributing **11/21 points**, **24% above** the overall **average of 8.86**.